2	**(a) (i)** N: (total) number of <u>molecules</u>	B1	[1]
	(ii) $<c^2>$: mean square speed/velocity	B1	[1]
	(b) $pV = ⅓Nm<c^2> = NkT$		
	(mean) kinetic energy = ½ $m<c^2>$	C1	
	algebra clear leading to ½ $m<c^2> = (3/2)kT$	A1	[2]
	(c) (i) *either* energy required = $(3/2) \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23}$	C1	
	= 12.5 J (*12J if 2 s.f.*)	A1	[2]
	or energy = $(3/2) \times 8.31 \times 1.0$	(C1)	
	= 12.5 J	(A1)	
	(ii) energy is needed to push back atmosphere/do work against atmosphere	M1	
	so total energy required is greater	A1	[2]